DEBRE TABOR UNIVERSITY

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Project Proposal Title: Office File Management System

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1. Introduction

1.1. Background

The proposed web-based **Office File Management System** is a digital archiving platform designed to modernize administrative workflows. It replaces physical paperwork with a secure, centralized web repository for letters and records. In the context of Ethiopia's growing digital economy, this system serves as a bridge between traditional paper filing and modern automated solutions, significantly improving document retrieval speed and data security.

1.2 Description of the Existing System

Currently, most local offices rely on a Traditional Manual System. Incoming and outgoing letters are hand-written in paper ledger books and stored in physical cabinets. To retrieve or share documents, staff must be physically present to search through paper archives. This manual process is time-consuming, requires massive physical storage space, and is highly prone to document wear-and-tear.

2. Statement of the Problem and Justification

2.1 Statement of the Problem

The traditional manual approach to archiving files in local offices suffers from the following critical bottlenecks:

1. **Time Inefficiency:** Searching for historical letters or archives manually through paper boxes and hand-written logbooks causes significant administrative delays.
2. **Loss of Documents:** Physical papers are vulnerable to tearing, wear-and-tear, or permanent misplacement.
3. **Lack of Remote Access:** Staff cannot search for or verify document metadata (like reference numbers or dates) unless they are physically present in the file archive room.
4. **Security Gaps:** Physical cabinets lack granular access controls, making it easy for unauthorized individuals to view confidential administrative correspondence.
5. **No Activity Tracking:** There is no mechanism to track who accessed, updated, or removed a specific file from the archives.

2.2. Justification

Developing this system is essential to modernize organizational workflows. By digitizing documents, we eliminate physical space constraints, protect records from physical damage, and ensure confidential information is secured through Role-Based Access Control (RBAC). Furthermore, it allows administrators to track user activities and retrieve any document in seconds, transforming the office into a highly transparent and productive environment.

3.Objectives of the Project

3.1. General Objective

To design and develop a web-based **Office File Management System** that automates document registration, file uploading, access control, and searching for office administrations.

3.2. Specific Objectives

- To create a secure User Authentication and Role-Based Access Control (RBAC) system for Admin and Staff.
- To develop an Admin Dashboard for managing user accounts, departments, and file categories.
- To build a File Upload module for registering document metadata and uploading PDFs/images.
- To implement an Advanced Search and Filtering mechanism for instant document retrieval.
- To integrate an In-App PDF Viewer to preview files without forcing downloads.
- To design a Soft Delete (Trash Bin) system to protect records from accidental deletion.
- To create an Activity Log (Audit Trail) to monitor and record user actions for security.

4. Scope of the Project

The scope of this project focuses on a secure, web-based platform accessible via desktop and tablet web browsers.

4.1 in-Scope (System Functional Requirements)

The system will perform the following core functions:

1. **User Login & Security:** Allows registered staff to log in securely using their username and password.
2. **Role-Based Access Control (RBAC):** Restricts system permissions based on user roles:
    - **Admin:** Can register users, manage departments, and restore deleted files.
    - **Staff:** Can only upload, search, and preview documents.
3. **Document Registration & Metadata Entry:** A form to register key details of a letter (Reference Number, Subject, Sender/Receiver, and Date).
4. **File Uploading:** Allows staff to upload scanned PDF or image copies of the letters and link them with the registered metadata.
5. **Department Categorization:** Allows files to be organized into virtual folders based on departments (e.g., HR, Finance, Logistics).
6. **Document Search:** Enables users to search for files instantly by Reference Number or Subject, and filter results by department or date.

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7. **Trash Bin (Soft Delete):** Moves deleted files to a temporary trash bin, allowing the Admin to restore them if deleted accidentally.
8. **Activity Logging (Audit Trail):** Automatically records which user uploaded, downloaded, or deleted which file, along with the timestamp.

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4.2. Out-of-Scope

1. Native mobile applications (Android/iOS apps).
2. Optical Character Recognition (OCR) for converting scanned text into editable text.
3. Offline desktop versions or integration with third-party external cloud storage APIs (e.g., Google Drive API).

5. Methodology and Tools

5.1. Methodology: Agile

We selected the **Agile-Scrum** software development model because our project timeline is limited. This model allows us to deliver a functional system quickly through iterative steps. It also provides the flexibility to show early prototypes to our advisor and make improvements based on feedback.

The development process will follow these phases:

Phases: Requirement Analysis => System Design => development (Coding) => Testing => Deployment

5.2. Tools and Technologies

The following development tools and technologies will be used to build the system:

- **Frontend:** HTML5, CSS3, JavaScript, and Bootstrap (to ensure the system is responsive on desktops and tablets).
- **Backend:** PHP (to handle server-side logic, file uploading, and database connections).
- **Database:** MySQL (relational database to store user credentials and document metadata securely).
- **Local Server Environment:** XAMPP (to run Apache and MySQL locally).

6.Schedule of the Project (Sample Timeline)

- **Phase 1: Requirement Analysis and System Design (UI/UX & Database Schema)** - Week 1
- **Phase 2: Database Setup, User Authentication, & File Upload Module** - Week 2
- **Phase 3: Frontend Integration, Search Engine, & Advanced Features (Trash Bin, Activity Log)** - Week 3
- **Phase 4: System Testing, Bug Fixing, Documentation, & Final Submission** - Week 4